BITTU KUMAR

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CAREER OBJECTIVE

Computer Science Engineering student specializing in Artificial Intelligence and Machine Learning, with hands-on project experience in AI-based fraud analysis, IoT monitoring, embedded systems, and web application development. Interested in building practical technology solutions, conducting research, and applying data-driven approaches to real-world problems.

EDUCATION

B.Tech CSE (AI & ML)

Sandip University, Sijoul, Madhubani

TECHNICAL SKILLS

Programming	Python (Intermediate), C , Embedded C/C++
AI, Data & Visualization	Fraud detection concepts, Pandas, NumPy, NetworkX, Plotly, PyVis
Hardware & Embedded	Arduino, ESP32, Raspberry Pi
IoT & Prototyping	Sensor interfacing, real-time monitoring, Wokwi, Arduino IDE
Web Development	HTML, CSS, JavaScript; Streamlit; Flask (project-specific use)
Tools & Version Control	Git, GitHub, VS Code
Data & Databases	SQL, MongoDB, Pandas, NumPy
CS Fundamentals	Data Structures, Algorithms, Problem Solving

PROJECTS

FraudShield AI | AI-Based Fraud Detection Dashboard

Developed a Python and Streamlit-based dashboard to analyse transaction data, flag suspicious activity, assess risk indicators, and visualise relationships between accounts using data-processing and network-visualisation tools.

Smart Fisheries Monitoring System

Prototyped an IoT monitoring system using an ESP32, sensors, and output devices to display environmental readings and support alerts/automation. Wokwi was used for simulation; pH input in the prototype should be described as simulated if implemented with a potentiometer.

AgriScan AI Pro

Designed an AI- and IoT-based smart agriculture system concept for monitoring crop and field conditions. The research architecture explores Raspberry Pi 5, ESP32, computer vision, sensor fusion, and crop-quality assessment.

AI Chatbot | Saarthi AI

Developed an AI-powered chatbot for answering user queries through a web interface. Resume project details identify Python, Flask, and AI APIs as the technology stack.

Portfolio Builder | PortfolioFlow

Created a responsive portfolio-builder website concept to help students and professionals present their work using modern templates.

Research Paper

AgriScan AI | AI-Based Crop Quality Grading and Post-Harvest Decision Support

Developing a multi-sensor AI framework using Raspberry Pi 5, ESP32, computer vision, and sensor fusion to assess crop quality, moisture, freshness, and defects in real time. Integrating AI-based image analysis and sensor data to provide crop grading, storage recommendations, and post-harvest decision support for smarter agriculture.

Status: Manuscript written; not yet published.

CERTIFICATIONS

- VICODATHON by AB TALKS
- AIML FOR Geodata Analytics (indian institution of Remote Sensing (IIRS)) Indian Space Research Organization (ISRO)
- Introduction to AI
- HackerRank – Python (Basic) Certificate

SOFT SKILLS

• Research & Critical Thinking • Teamwork & Collaboration • Problem Solving • Leadership • Time Management • Creativity & innovation
• Project Management • Presentation Skills